

Atmospheric AI Model Benchmark & Verification Metric Handbook

Evaluation Protocols for Physics-Informed Neural Networks, GraphCast, ConvLSTM, and Probabilistic Convective Nowcasting Verification

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1. Paradigm Shift to Deep Learning in Meteorology

Physics-Informed Deep Learning architectures (GraphCast, FuXi, Pangu-Weather, and ConvLSTM) leverage 40+ years of ECMWF ERA5 and IMD hourly reanalysis to learn non-linear atmospheric state transitions at fractions of traditional numerical computation cost (0.1s on a single GPU vs 2 hours on a 10,000-core supercomputer).

2. Standard Verification Metrics for Categorical Precipitation

Deterministic point-wise accuracy metrics like Mean Squared Error (MSE) fail in extreme precipitation forecasting because minimizing MSE produces blurred, over-smoothed fields. Forecasts must be validated using threshold contingency matrices:

Metric Name	Formula / Definition	Ideal Value	Operational Utility
Critical Success Index (CSI)	Hits / (Hits + False Alarms + Misses)	1.00 (100%)	Evaluates extreme localized rain events
Fractions Skill Score (FSS)	$1 - (\text{MSE}_f / \text{MSE}_{\text{ref}})$	1.00 (>0.5 useful)	Spatial displacement scale tolerance
Equitable Threat Score (ETS)	$(\text{Hits} - \text{Expected}) / (\text{Hits} + \text{FA} + \text{Miss} - \text{Exp})$	> 0.40	Accounts for random chance correct hits
Probability of Detection (POD)	Hits / (Hits + Misses)	1.00	Warning reliability for flash flood alerts

3. Physics-Informed Loss Constraints (PINNs)

```
# PyTorch Loss Function Enforcing Atmospheric Mass Conservation
def physics_informed_loss(pred_u, pred_v, pred_rho, true_precip):
    mse_loss = F.mse_loss(pred_precip, true_precip)
    div_uv = torch.gradient(pred_u)[0] + torch.gradient(pred_v)[1]
    continuity_loss = torch.mean(torch.abs(div_uv)) # V·V = 0 incompressible limit
    return mse_loss + 0.15 * continuity_loss
```

Summary: Embedding Navier-Stokes divergence penalties directly into neural network backpropagation eliminates unphysical artifact generation and ensures mass balance across complex Himalayan terrain.